

The equation $x^4 + xy + y^3 = 1$

A finiteness result and the obstruction to infinitude

Answer. The proposed infinitude statement is false. In fact, there are only finitely many rational solutions, and therefore only finitely many integer solutions.

Scope. The algebraic checks below are proved explicitly. The final deduction invokes the genus formula for a smooth plane curve and Faltings' theorem, both stated where used. Thus the conclusion is unconditional, but this is *not a fully self-contained proof of finiteness*. A proof of the requested infinitude cannot be supplied because that assertion is false. No claim of novelty is made.

1. Intuition: check the curve before seeking a family

The dominant terms suggest the balance $y^3 \approx -x^4$. This describes the possible size of a large solution; it does not produce exact integer solutions. Before attempting a recurrence or parametrization, the useful structural question is whether the curve is singular. Singularities can reduce the genus of a plane curve and sometimes permit a rational parametrization. Here the degree is four, and all points turn out to be smooth, so the genus is three. That is the decisive obstruction.

2. Explicit geometric checks

Put $f(x, y) = x^4 + xy + y^3 - 1$. Its projective closure C is the curve

$$F(X, Y, Z) = X^4 + XYZ^2 + Y^3Z - Z^4 = 0. \quad (1)$$

All smoothness and irreducibility checks are made over $\mathbb{C}$, so they also cover nonreal points.

The point at infinity is smooth. Setting $Z = 0$ in (1) gives $X^4 = 0$. Hence the only point at infinity is $P = [0 : 1 : 0]$. Since

$$F_Z = 2XYZ + Y^3 - 4Z^3, \quad F_Z(0, 1, 0) = 1,$$

P is nonsingular.

The curve is geometrically irreducible. Suppose that $F = GH$ over $\mathbb{C}$, where G, H are homogeneous of positive degrees d and $4 - d$. Restricting to $Z = 0$ gives

$$G(X, Y, 0)H(X, Y, 0) = X^4.$$

Neither restricted factor is zero. Since their product is a power of X , homogeneity forces

$$G(X, Y, 0) = cX^d, \quad H(X, Y, 0) = c^{-1}X^{4-d} \quad (c \neq 0).$$

Consequently $G(P) = H(P) = 0$. The product rule would then give $\nabla F(P) = G(P)\nabla H(P) + H(P)\nabla G(P) = 0$, contradicting $F_Z(P) = 1$. Thus F has no such factorization.

There are no affine singularities. An affine singularity would satisfy

$$f = 0, \quad f_x = 4x^3 + y = 0, \quad f_y = x + 3y^2 = 0. \quad (2)$$

The case $x = 0$ forces $y = 0$, but $f(0, 0) = -1$, so $x \neq 0$. Substituting $y = -4x^3$ into the last equation yields

$$x + 48x^6 = 0, \quad x^5 = -\frac{1}{48}.$$

Also (2) gives $xy = -4x^4$ and $y^3 = -xy/3 = 4x^4/3$. Hence

$$0 = f = x^4 - 4x^4 + \frac{4}{3}x^4 - 1 = -\frac{5}{3}x^4 - 1, \quad x^4 = -\frac{3}{5}.$$

Taking the ratio of the two nonzero powers forces

$$x = \frac{x^5}{x^4} = \frac{5}{144}.$$

This is a positive real number and cannot have fourth power $-3/5$. The contradiction excludes every affine singularity, including complex ones. Together with the check at P , it proves that C is smooth.

3. Why the solution set is finite

We now use the standard *genus formula*: a smooth, geometrically irreducible projective plane curve of degree d in characteristic zero has genus $(d-1)(d-2)/2$. The checks above justify its application and give

$$g(C) = \frac{(4-1)(4-2)}{2} = 3.$$

Faltings' theorem (invoked). *If a smooth, projective, geometrically irreducible curve over a number field K has genus at least two, its set of K -rational points is finite.*

Theorem 1. *The equation $x^4 + xy + y^3 - 1 = 0$ has only finitely many rational solutions. In particular, it has only finitely many integer solutions.*

Proof. The curve C is defined over $\mathbb{Q}$, is smooth and geometrically irreducible, and has genus three. Faltings' theorem therefore makes $C(\mathbb{Q})$ finite. Each rational solution determines a point $[x : y : 1] \in C(\mathbb{Q})$, and different pairs determine different points. Thus the rational solution set is finite. Its subset of integer solutions is finite as well. $\square$

4. Explicit solutions and the limit of this argument

Five integer solutions, verified by direct substitution, are

$$\boxed{(-1, -1), \quad (-1, 0), \quad (-1, 1), \quad (0, 1), \quad (1, 0).}$$

They are exactly the solutions with $|x| \leq 1$: for $x = -1, 0, 1$, respectively, the equation becomes $y^3 - y = 0$, $y^3 = 1$, and $y^3 + y = 0$.

The proof above does *not* establish that these five points form the complete integer solution set, and it supplies no explicit bound on the coordinates. Neither is needed to refute infinitude. The precise result proved here is finiteness, with its reliance on the two stated general theorems made explicit.